

Lecture 11

Quadratic Constraints for Analyzing Stochastic Gradient Descent

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In this lecture, we study the stochastic gradient descent (SGD) method for the following finite-sum optimization problem

$$\underset{x \in \mathbb{R}^p}{\text{minimize}} \quad f(x) := \frac{1}{n} \sum_{i=1}^n f_i(x), \quad (11.1)$$

where $f : \mathbb{R}^p \rightarrow \mathbb{R}$ is a strongly convex objective function. SGD is the baseline algorithm for large-scale machine learning [1]. We will discuss how SGD can be represented as a feedback dynamical system, as shown in Figure 11.1, and how the dissipation inequality approach from earlier lectures can be tailored to analyze it. We will show that under typical assumptions, SGD with constant stepsize converges linearly to a neighborhood around the optimal point, and the stepsize trades off the convergence speed against the steady-state accuracy.

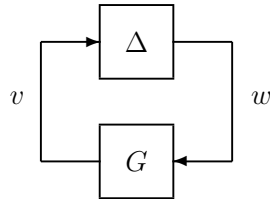


Figure 11.1. The Block-Diagram Representation for Feedback Interconnection $F_u(G, \Delta)$

11.1 Motivations: ERM and Supervised Learning

Empirical risk minimization (ERM) is a key paradigm in machine learning and naturally leads to the finite-sum problem (11.1). Many supervised learning tasks, including ridge regression, logistic regression, and support vector machines, can be formulated as

$$\underset{x \in \mathbb{R}^p}{\text{minimize}} \quad f(x) := \frac{1}{n} \sum_{i=1}^n \ell_i(x) + \gamma \Omega(x),$$

which is (11.1) with $f_i(x) = \ell_i(x) + \gamma \Omega(x)$. Here x parameterizes a prediction/classification model, and the training set consists of n data points. The loss $\ell_i(x)$ measures how well x performs on the i -th data point, and $\Omega(x)$ is some sort of regularizer that is used to introduce useful prior bias for the learning.

Example 1: Ridge regression. Given data points $a_i \in \mathbb{R}^p$ and $b_i \in \mathbb{R}$, ridge regression fits a linear model $b \approx a^\top x$ by minimizing

$$f(x) = \frac{1}{n} \sum_{i=1}^n (a_i^\top x - b_i)^2 + \frac{\gamma}{2} \|x\|^2. \quad (11.2)$$

This is a positive definite quadratic function with Hessian $\frac{2}{n} \sum_{i=1}^n a_i a_i^\top + \gamma I \succ 0$. Hence f is m -strongly convex and L -smooth (why?).

Example 2: ℓ_2 -regularized logistic regression. Given features $a_i \in \mathbb{R}^p$ and labels $b_i \in \{-1, 1\}$, logistic regression fits a linear classifier that predicts $b = 1$ when $a^\top x \geq 0$ and $b = -1$ otherwise. For example, a_i can stack the pixels of an image, with $b_i = 1$ for a cat and $b_i = -1$ for a dog. The objective function is

$$f(x) = \frac{1}{n} \sum_{i=1}^n \log \left(1 + e^{-b_i a_i^\top x} \right) + \frac{\gamma}{2} \|x\|^2, \quad (11.3)$$

which is also L -smooth and m -strongly convex.

Finite-sum structure of ERM. The gradient method evaluates $\nabla f = \frac{1}{n} \sum_{i=1}^n \nabla f_i$ at every iteration. In other words, the gradient on all the data points is needed, and the cost of each iteration scales as $O(n)$. For big data applications, n is very large and this cost is high. This motivates stochastic methods that sample one or a small batch of data points to estimate the gradient at every iteration, so that the per-iteration cost does not depend on n . The hope is that there is a lot of redundancy between data points, and these methods will work well in some average sense in the long run.

11.2 Background on SGD

SGD iterates as

$$x_{k+1} = x_k - \alpha \nabla f_{i_k}(x_k), \quad (11.4)$$

where the index i_k is sampled uniformly from $\{1, 2, \dots, n\}$, independently of the past indices $i_0, \dots, i_{k-1}$. Compared with the gradient method $x_{k+1} = x_k - \alpha \nabla f(x_k)$, only one data point is used at each step, and hence the per-iteration cost is independent of n . Mini-batch SGD averages the gradients over a small sampled batch; for simplicity, we focus on the single-sample version (11.4). Throughout this lecture, x_0 is deterministic, and $\mathbb{E}$ denotes the expectation with respect to the random indices.

The stochastic gradient is an unbiased estimate. Notice x_k is completely determined by x_0 and $i_0, \dots, i_{k-1}$, and i_k is independent of these indices. Therefore, if we fix x_k and average over the n equally likely values of i_k , we obtain

$$\mathbb{E}[\nabla f_{i_k}(x_k) | x_k] = \frac{1}{n} \sum_{i=1}^n \nabla f_i(x_k) = \nabla f(x_k). \quad (11.5)$$

Roughly speaking, SGD behaves like the gradient method in some average sense.

The optimal point may not be a fixed point. Notice $\nabla f(x^*) = 0$ does not mean $\nabla f_i(x^*) = 0$ for every i . Hence SGD will not stay at the optimal point x^* even if it is initialized there: $x_1 = x^* - \alpha \nabla f_{i_0}(x^*) \neq x^*$ whenever $\nabla f_{i_0}(x^*) \neq 0$. As a result, SGD with a constant stepsize only converges to some tolerance of the optimum. If a diminishing stepsize is used, SGD converges to the optimum, but only at a sublinear rate.

11.3 SGD as a Feedback System

Recall that the gradient method is the feedback interconnection $F_u(G, \Delta)$ of an LTI system G and the static nonlinearity $\Delta = \nabla f$. SGD has the same form if we keep G and let Δ depend on the random index i_k :

$$\begin{aligned} \xi_{k+1} &= A\xi_k + Bw_k, \\ v_k &= C\xi_k, & A = I, \quad B = -\alpha I, \quad C = I. \\ w_k &= \nabla f_{i_k}(v_k), \end{aligned} \quad (11.6)$$

To see this, we just set $\xi_k = x_k$. Then $v_k = x_k$ and $x_{k+1} = x_k - \alpha \nabla f_{i_k}(x_k)$, which is exactly (11.4). Since only the perturbation Δ changes, it is not surprising that the convergence proofs for the gradient method and SGD are quite similar. The dissipation inequality approach can directly handle the stochastic mapping ∇f_{i_k} , as long as we can construct informative quadratic constraints (supply rates) for it. Such quadratic constraint and supply rates will only hold in the average sense, i.e., they take the form

$$\mathbb{E} \begin{bmatrix} \xi_k - \xi^* \\ w_k \end{bmatrix}^T X \begin{bmatrix} \xi_k - \xi^* \\ w_k \end{bmatrix} \leq c \quad (11.7)$$

with well-chosen X and c .

11.4 Convergence of SGD with a Constant Stepsize

Dissipation inequality in expectation. Suppose $\xi_{k+1} - \xi^* = A(\xi_k - \xi^*) + Bw_k$, and the supply rate satisfies

$$S_k := \begin{bmatrix} \xi_k - \xi^* \\ w_k \end{bmatrix}^T X \begin{bmatrix} \xi_k - \xi^* \\ w_k \end{bmatrix}, \quad \mathbb{E} S_k \leq c$$

for some constant $c \geq 0$. If there exists $P \succ 0$ such that

$$\begin{bmatrix} A^\top P A - \rho^2 P & A^\top P B \\ B^\top P A & B^\top P B \end{bmatrix} - X \preceq 0, \quad (11.8)$$

then, exactly as in the deterministic case, $V(\xi) := (\xi - \xi^*)^\top P (\xi - \xi^*)$ satisfies the dissipation inequality $V(\xi_{k+1}) \leq \rho^2 V(\xi_k) + S_k$ for every realization of the random indices. Since the supply rate is only controlled on average, we take expectations and iterate:

$$\mathbb{E}V(\xi_{k+1}) \leq \rho^2 \mathbb{E}V(\xi_k) + c \implies \mathbb{E}V(\xi_k) \leq \rho^{2k} V(\xi_0) + \frac{c}{1 - \rho^2}. \quad (11.9)$$

So the method converges linearly to a ball whose size is controlled by $\frac{c}{1 - \rho^2}$. If $c > 0$, the stochastic input keeps on delivering energy to the system, and the internal energy may not converge to 0. If $c = 0$, we recover the deterministic analysis used for the gradient method.

Assumptions. We assume that f is m -strongly convex and each f_i is L -smooth and convex. Then f has a unique minimizer x^* , and $\nabla f(x^*) = 0$. Only the average f needs to be strongly convex. Notice that L is a common smoothness constant for all f_i , which can be larger than that of f ; for ridge regression (11.2), we can choose $L = \max_i (2\|a_i\|^2 + \gamma)$. For SGD, $\xi^* = x^*$, and the error dynamics is $x_{k+1} - x^* = (x_k - x^*) - \alpha w_k$. We emphasize that we do not shift w_k to $w_k - w^*$: the input at the optimum is $\nabla f_{i_k}(x^*)$, which is random and typically nonzero. Again, we perform our analysis in two steps: construct the supply rates, and then solve an LMI.

11.4.1 Step 1: Supply Rates

Define $\sigma^2 := \frac{1}{n} \sum_{i=1}^n \|\nabla f_i(x^*)\|^2$, which measures how much the individual gradients disagree at the optimum. With $z_k := [(x_k - x^*)^\top, w_k^\top]^\top$, we will show

$$\mathbb{E} [z_k^\top X_1 z_k] \leq 2\sigma^2 =: M, \quad X_1 := \begin{bmatrix} 0 & -LI \\ -LI & I \end{bmatrix}, \quad (11.10)$$

$$\mathbb{E} [z_k^\top X_2 z_k] \leq 0, \quad X_2 := \begin{bmatrix} 2mI & -I \\ -I & 0 \end{bmatrix}. \quad (11.11)$$

Notice that it is the first supply rate that causes the convergence issue for SGD: it keeps on delivering energy to the system. Both inequalities only hold in the average sense; see also [2]. In both proofs, we first fix x_k and average over i_k using (11.5), and then take the total expectation.

Proof of the second inequality. We have $z_k^\top X_2 z_k = 2m\|x_k - x^*\|^2 - 2(x_k - x^*)^\top w_k$. Using (11.5) and $\nabla f(x^*) = 0$, its conditional expectation given x_k is

$$2m\|x_k - x^*\|^2 - 2(x_k - x^*)^\top (\nabla f(x_k) - \nabla f(x^*)) \leq 0,$$

where the inequality follows from adding the strong convexity inequalities at x_k and x^* . A single sample does not satisfy this bound in general, since f_{i_k} need not be strongly convex or minimized at x^* .

Proof of the first inequality. We have $z_k^\top X_1 z_k = \|w_k\|^2 - 2L(x_k - x^*)^\top w_k$. We decompose the stochastic gradient as

$$w_k = \underbrace{\nabla f_{i_k}(x_k) - \nabla f_{i_k}(x^*)}_{=: d_k} + \nabla f_{i_k}(x^*). \quad (11.12)$$

Since f_{i_k} is convex and L -smooth, the co-coercivity inequality (the sector bound with $m = 0$) gives $\|d_k\|^2 \leq L(x_k - x^*)^\top d_k$, no matter which index is sampled. Combining this with $\|a + b\|^2 \leq 2\|a\|^2 + 2\|b\|^2$ leads to

$$\|w_k\|^2 \leq 2L(x_k - x^*)^\top d_k + 2\|\nabla f_{i_k}(x^*)\|^2.$$

Now we fix x_k and average over i_k . Since $\nabla f(x^*) = 0$, we have $\mathbb{E}[d_k | x_k] = \nabla f(x_k) = \mathbb{E}[w_k | x_k]$ and $\mathbb{E}[\|\nabla f_{i_k}(x^*)\|^2 | x_k] = \sigma^2$. Hence $\mathbb{E}[\|w_k\|^2 | x_k] \leq 2L(x_k - x^*)^\top \mathbb{E}[w_k | x_k] + 2\sigma^2$. Rearranging and taking the total expectation gives (11.10).

11.4.2 Step 2: The LMI and the Convergence Bound

We apply (11.8) with $X = \lambda_1 X_1 + \lambda_2 X_2$, where $\lambda_1, \lambda_2 \geq 0$. By (11.10) and (11.11), the supply rate $S_k = \lambda_1 z_k^\top X_1 z_k + \lambda_2 z_k^\top X_2 z_k$ satisfies $\mathbb{E} S_k \leq \lambda_1 M$, i.e., $c = \lambda_1 M$. For simplicity, we choose $P = I$. Since $A = I$ and $B = -\alpha I$, every block in (11.8) is a scalar multiple of I , and the LMI reduces to the 2×2 condition

$$\begin{bmatrix} 1 - \rho^2 & -\alpha \\ -\alpha & \alpha^2 \end{bmatrix} - \lambda_1 \begin{bmatrix} 0 & -L \\ -L & 1 \end{bmatrix} - \lambda_2 \begin{bmatrix} 2m & -1 \\ -1 & 0 \end{bmatrix} \preceq 0. \quad (11.13)$$

Setting the left side to be a zero matrix gives

$$\lambda_1 = \alpha^2, \quad \lambda_2 = \alpha(1 - L\alpha), \quad \rho^2 = 1 - 2m\lambda_2 = 1 - 2m\alpha + 2mL\alpha^2. \quad (11.14)$$

We need $\lambda_2 \geq 0$, which requires $\alpha \leq \frac{1}{L}$. For $0 < \alpha < \frac{1}{L}$, we also have $\rho^2 < 1$, and (11.9) with $c = \lambda_1 M = 2\alpha^2\sigma^2$ and $1 - \rho^2 = 2m\alpha(1 - L\alpha)$ gives

$$\mathbb{E}\|x_k - x^*\|^2 \leq (1 - 2m\alpha + 2mL\alpha^2)^k \|x_0 - x^*\|^2 + \frac{\alpha\sigma^2}{m(1 - L\alpha)}. \quad (11.15)$$

We write this as $\mathbb{E}\|x_k - x^*\|^2 \leq \rho^{2k}\|x_0 - x^*\|^2 + H$. Here $\rho^2 = 1 - 2m\alpha + O(\alpha^2)$ quantifies the convergence speed, and $H = \frac{\lambda_1 M}{1 - \rho^2} = O(\alpha)$ quantifies the accuracy.

11.5 Interpreting the Bound

Speed versus accuracy. The bound (11.15) has a transient term, which decays linearly as for the gradient method, and an error floor H , which does not decay: SGD with a constant stepsize only converges to a ball around x^* . A smaller α gives a smaller error floor, $H \approx \frac{\sigma^2}{m}\alpha$,

but a slower transient, $1 - \rho^2 \approx 2m\alpha$. Roughly speaking, halving α halves the error floor but doubles the number of iterations needed for the transient to die out. The fastest certified transient is obtained at $\alpha = \frac{1}{2L}$, which gives $\rho^2 = 1 - \frac{m}{2L}$ and $H = \frac{\sigma^2}{mL}$; a larger α makes both terms worse in our bound. This is the fundamental trade-off for SGD with a constant stepsize: one cannot make the convergence fast and the final error small at the same time. It is one motivation for the common practice of decreasing the stepsize (learning rate) in stages during training.

Where does the error floor come from? Using (11.12), we can rewrite SGD as

$$x_{k+1} = x_k - \alpha (\nabla f_{i_k}(x_k) - \nabla f_{i_k}(x^*)) - \alpha \nabla f_{i_k}(x^*).$$

The first input term satisfies the co-coercivity constraint for every realization of i_k , so it does not deliver energy into the system. The troublesome term is $\nabla f_{i_k}(x^*)$. It has zero mean, since $\frac{1}{n} \sum_{i=1}^n \nabla f_i(x^*) = \nabla f(x^*) = 0$, but its second moment is σ^2 . It does not depend on the state at all, and it keeps on delivering energy into the system. This is exactly where $M = 2\sigma^2$ in (11.10) comes from. For example, if $x_0 = x^*$, then $\mathbb{E}\|x_1 - x^*\|^2 = \alpha^2 \sigma^2$. This energy viewpoint also explains why $H = O(\alpha)$: the expected energy supplied at each step is at most $\lambda_1 M = 2\alpha^2 \sigma^2$, the fraction of the stored energy dissipated at each step is $1 - \rho^2 \approx 2m\alpha$, and the error floor is the level at which the two balance.

When is there no error floor? If $\nabla f_i(x^*) = 0$ for all i , i.e., all the f_i share the same minimizer, then $\sigma^2 = 0$, and SGD with a constant stepsize converges linearly to x^* . An example is least squares with $f_i(x) = (a_i^\top x - b_i)^2$, where the vectors a_i span $\mathbb{R}^p$ and the data can be fit exactly, i.e., $a_i^\top x^* = b_i$ for all i .

Diminishing stepsizes. Since the energy supplied at step k is at most $2\alpha_k^2 \sigma^2$, a diminishing stepsize removes the error floor and can give a $O(1/k)$ convergence rate for the optimal point. The details are omitted here.

References

- [1] L. Bottou. Large-scale machine learning with stochastic gradient descent. In *Proceedings of COMPSTAT'2010*, pages 177–186. Springer, 2010.
- [2] B. Hu, P. Seiler, and L. Lessard. Analysis of biased stochastic gradient descent using sequential semidefinite programs. *Mathematical Programming*, 187(1–2):383–408, 2021.